Studying the Impact of Explainable AI in Digital Agriculture Solutions

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ABSTRACT

Although agriculture has traditionally appeared to be a perpetual industry, it has encountered a gradually rising set of significant challenges in recent years, including the alarming depletion of natural resources, the rising water and food demands, and the limited amount of arable land. In response to these challenges, the agricultural sector has seen an increased effort to embrace the digital revolution, harnessing emerging technologies to optimize sustainable agricultural processes and provide valuable support for informed decision-making.

Given the persistent lack of transparency in machine learning models, farmers' perceived complexity and low value of smart agriculture solutions, this research focuses on studying the impact of *explainable AI* techniques and multimodal data on farmers' user experience in digital agriculture solutions. In this context, we propose a novel collaborative platform, *AgriUXE*, particularly tailored for AI-driven digital agriculture applications. The platform will focus on augmenting the explainability of both captured multimodal data and machine learning models' predictions. Moreover, it is crucial to evaluate how an optimized user experience, achieved through the development of transparent data-driven solutions in collaboration with key farm stakeholders, influences the expectations of small and medium-sized farmers regarding smart farming technologies.

CCS CONCEPTS

• **Applied computing** → **Agriculture**; • **Computing methodologies** → **Machine learning**; • **Human-centered computing** → **User centered design**; **Human computer interaction (HCI)**.

KEYWORDS

Digital Agriculture, Explainable AI, Machine Learning, Multimodal Systems, Human-Computer Interaction

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1 INTRODUCTION AND BACKGROUND

According to the *Food and Agriculture Organization*, the world population is expected to grow to numbers between 9.4 and 10.1 billion by 2050 with a certainty confidence of 95% [6, 10]. To meet the food demand from this population growth, agricultural production will need to increase by around 70% compared to 2009[7]. In order to fulfill these food expectations and address the need for sustainable solutions amid emerging global challenges, such as those posed by the recent climate crisis, the agricultural sector plays a key role. Immediate action and investment in agriculture have the potential to tackle these challenges by providing practical and efficient solutions [8].

Particularly, recent progress in the digitalization of agriculture, characterized by the adoption of emerging technologies and innovative services in agriculture [4], has the potential to address issues such as weed and plant disease detection, carbon sequestration, crop yield estimation, and reliable food systems. Emerging smart farming technologies provide key farm stakeholders with sustainable data-driven solutions by enabling a holistic perspective on farms and their internal processes.

The interest in revolutionizing the agricultural sector with technological solutions has given rise to multiple concepts like Digital Agriculture, Agriculture 4.0, and Precision Agriculture. Despite the terminology used, all these terms imply a digitalization effort of farm processes. Based on data sources such as sensors and satellite imagery, smart farming technologies can provide farmers with data-driven predictions, enabling them to make more efficient decisions [13].

Among the applications of digital technologies within agriculture, the existing research literature highlights the potential of employing technologies for crop yield prediction [2, 5], market linkage (i.e., accessing agricultural markets) [11], plant disease and weed detection [9, 15], livestock tracking [19, 20], optimizing crop breeding processes [18], policy (e.g., how to tackle data democratization in farm settings) [1], and efficient food systems [14]. Concerning the latter, particular efforts have extended beyond national boundaries, exemplified by initiatives like the European Commission's *Farm to Fork* strategy. This strategy aims to transition food systems toward comprehensive monitoring and rigorous testing across all processes, encompassing food traceability, integrity, safety, and security [3].

However, the core of agriculture practices still heavily relies on traditional methods, with farmers, especially small and medium-scale farmers, not significantly adopting digital technologies in their farms [12]. Complex interactions, widespread technological

illiteracy among farmers, and high investment costs are among the set of barriers preventing a quicker adoption of new technologies on farm settings [12, 17]. Furthermore, given the inherent potential of AI systems, a significant number of digital solutions involve employing machine learning models. Nonetheless, the increasing complexity of these models, often perceived as complete opaque "black boxes", especially in deep learning contributions, hinders the development of transparent and explainable predictive solutions.

Therefore, this thesis aims to research and frame a novel digital platform that incorporates multimodal data and *explainable AI* techniques to offer explainable farm data-driven insights. By improving both data explainability and the interpretability of machine learning predictions, we seek to study, in collaboration with key farm stakeholders, how explainability impacts farmers' user experience in digital agriculture applications.

Considering the digital adoption gap prevalent among small and medium-scale farmers, this work will also focus in investigating how an optimized user experience in smart agriculture solutions can influence farmers' expectations regarding smart farming technologies.

2 RESEARCH QUESTIONS

From the established research context and motivation detailed in previous sections, the following research questions have arisen:

RQ₁ How does the integration of XAI-based techniques influences user experience in digital agriculture applications, and consequently impact the adoption of smart farming technologies among small and medium-size farmers?

RQ₂ How can AI be applied on farms that handle multimodal data collected from heterogeneous sources?

RQ₃ How should a collaborative platform be designed to maximize farm support in the common interest?

The stated research questions can be condensed in the subsequent research statement: "We propose the integration of XAI-based models and tools into digital agriculture applications to assess their influence on farmers' user experience, aiming towards the adoption of smart farming technologies by small and medium-size farmers".

3 EXPECTED CONTRIBUTIONS

Based on the vision and existing challenges identified in the associated research context, we ought to answer the proposed questions by extending previous research work. Through this process, we expect to expand the research domain with the following main contributions:

- **AgriUXE Platform**: The main contribution of the proposed work lies in the design and development of an explainable multimodal platform named *AgriUXE* (Agricultural eXperience Enhanced through eXplainability). *AgriUXE* aims to enhance farmers' user experience in digital agricultural applications by augmenting the explainability of both multimodal data and machine learning models' predictions. This entails integrating XAI novel techniques, design principles, and architectural guidelines adhering to this paradigm. The platform will be tailored for AI-driven digital agriculture applications, providing explanations for both the captured multimodal data and the inference process. The latter will encompass insights from individual machine learning predictions (explaining the rationale behind decisions) as well as a consolidated multimodal prediction explanation derived through a late fusion technique combining the individual unimodal predictions. The platform is intended to be collaborative as it will enable data collection from not only individual farms as well as shared pools of farm data from multiple farms connected to it, allowing a broader obtained knowledge that aims to enrich involved stakeholders. By guiding farmers toward transparent data-driven decisions, we further intend to study how enhanced user experience influences farmers' expectations of digital agriculture technologies. By doing so, we hope to propose initial contributions towards increasing the adoption rates of smart farming technologies in small and medium-size farms.
- **Proofs of Concepts and Prototypes**: The development approach will involve integrating *Proof of Concepts* to rigorously test the solution against defined evaluation criteria. While encompassing generic experiments, there should also be a focus on targeted feature evaluations, such as assessing the usability of the developed XAI models. If any of the experiments introduces pertinent breakthroughs, they will be considered for future publications.
- **Usability and User Experience Assessments**: We will conduct user experience evaluations through comprehensive user studies, employing standard data collection procedures such as observation, interviews, and user questionnaires. Combined with consistently gathering feedback from end users across development stages, we will integrate insights from these diverse user studies towards creating a tailored solution.

4 WORK PLAN

This work started in 2022 and is currently in its second year, expected to last until 2026. The work plan aligns with the scheduling and checkpoints devised in the Doctoral Program at NOVA SST, organized around three primary stages: preparation, research, and conclusion.

The research is being conducted under the supervision of Prof. Rui Neves Madeira (ORCID: 0000-0001-7360-3855) and Prof. Pedro Albuquerque Santos (ORCID: 0000-0001-8159-1364), whose significant scientific contributions to the field of multimodal systems endorses sound scientific research.

4.1 Past Work and Work In Progress

The preparation phase lasted 12 months and focused on defining the research trajectory. To solidify the premises for the research problem, an extensive literature review was conducted. The objective was to understand the current state-of-the-art literature, encompassing not only the barriers and expectations of farmers but also the applications of relevant subjects like AI, XAI, multimodal systems, and HCI in the field of digital agriculture. This endeavor resulted not only in writing the thesis plan document, which has enabled a comprehensive structuring of the motivation and novelty of the Ph.D. work, but also in the proposal of an initial architecture of the *AgriUXE* platform (see Figure 1).

Regarding past related work, a vineyard integrated solution, named *AgriDash*, was introduced [16]. Tailored to the unique needs of small and medium-size farms, *AgriDash* offers vital information and recommendations to farmers through its capabilities of monitoring ecosystems, enabling visualization of multi-sensor data, and extracting insights from predictive models.

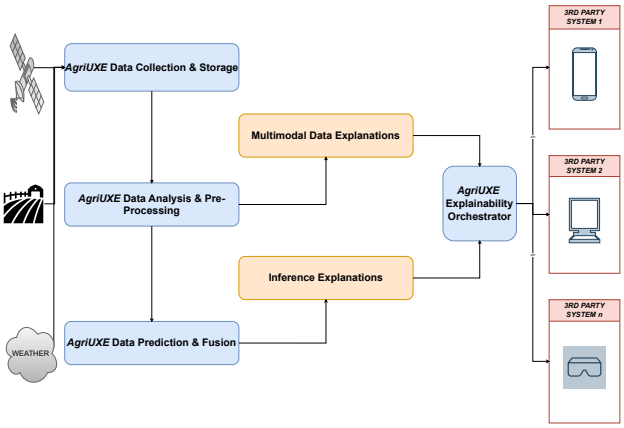

Figure 1: Simplified Proposal of *AgriUXE* architecture.

Within the context of ongoing work, the development of the *AgriUXE* platform has started and a first prototype is already underway, having *AgriDash* as a prototype starting point. This initial prototype aims to provide explainable insights to agronomists in the context of vineyard production, particularly, in yield estimation of grapes harvested for wine.

4.2 Future Work

The remaining 36 months of the Ph.D. will be focused on the research and conclusion stages. The research stage encompasses the development of the proposed solution, following an agile methodology. Through this iterative process, incremental prototypes are expected to be developed and tested by end users. Once initial stable versions of the product are established, our work will extend to conducting human factor evaluations. This phase peaks with the elaboration and discussion of the Ph.D. thesis proposal document. Regarding the final conclusion phase of the Ph.D., while conducting final user studies, it will be mostly dedicated to writing and defending the final dissertation document. Moreover, peer-reviewed publications will be written to relevant journals and conferences throughout the three stages of the Ph.D.

Furthermore, the platform development will be based on a co-creation methodology, involving all stakeholders in the whole process, especially farmers, as they will be the end users of the proposed solution. In this user-centered approach, it will be possible to involve all relevant knowledge actors besides the farmers, even including culture technicians, citizens, the civil society, in the co-creation process, applying open science practices, as much as possible.

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